Aravind Santhosh Kumar

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Education

Northeastern University Sept 2024 – Dec 2026
M.S. Computer Science *GPA: 4.0/4.0*

Vellore Institute of Technology Jun 2013 – May 2017
B.Tech. Computer Science

Skills

Languages: TypeScript, JavaScript, Python, Java
Backend: Node.js, NestJS, Express.js, FastAPI, REST APIs, Jest
Frontend: React, Next.js, Redux Toolkit
Databases: PostgreSQL, MySQL, Oracle DB, MongoDB
Cloud / Infrastructure: AWS Lambda, CloudWatch, Docker, NGINX, Redis, BullMQ, CI/CD, Git
Certification: AWS Certified Solutions Architect – Associate

Professional Experience

Software Development Engineer (Co-op) Remote – US Based
Geode *Jun 2025 – Dec 2025*

- Delivered discovery for 17,000+ dApps through sub-200ms Express.js APIs backed by PostgreSQL, supporting search, filtering, detailed views, and follow workflows.
- Cut third-party automation costs by 60% by replacing Make.com workflows with an in-house BullMQ/Redis ingestion pipeline; integrated LLM-generated summaries to enrich dApp metadata.
- Streamlined deep-link navigation from shortened URLs to the correct marketplace page, section, and pre-populated form state using an NGINX reverse-proxy and request-routing service.
- Cut deployment time by approximately 90% by introducing staging infrastructure and webhook-triggered CI/CD pipelines to replace manual deployments.

Software Developer 2 Bangalore, Karnataka, India
KPMG *Jun 2021 – Feb 2024*

- Led development of an enterprise Asset Lifecycle Management platform for a multinational energy-management company, building React/Redux interfaces and NestJS BFF services while mentoring junior developers.
- Reduced document-processing time by 15% by integrating Oracle Aconex via OAuth 2.0 for uploads, updates, versioning, bulk operations, and cross-platform synchronization across thousands of engineering documents.
- Built NestJS BFF integrations with AWS Lambda serverless backends and used CloudWatch to diagnose production errors and trace failures across application workflows.
- Eliminated a production frontend performance regression after API payloads grew from 10–20 records to hundreds by memoizing response data and removing redundant React re-renders.

Software Developer 2 Bangalore, Karnataka, India
Blume Global *Aug 2020 – Jun 2021*

- Built reusable React modules within a single-spa micro-frontend architecture and integrated Stripe payments, reducing transaction processing time by 40% compared with the previous manual invoicing workflow.

Software Developer Bangalore, Karnataka, India
Dsquare Solutions *Feb 2018 – Aug 2020*

- Enabled self-service analytics across MySQL, PostgreSQL, Google BigQuery, and CSV by building data-source integrations and dataset configuration workflows for InsightView using React, Redux, and Java (Play Framework).
- Expanded InsightView's analytics capabilities by developing reusable D3.js visualizations, configurable dashboard widgets, and PDF export workflows while working directly with clients to translate reporting requirements into product features.

Selected Projects

Arcanum: The Ten Trials | *React, FastAPI, PostgreSQL, Supabase, Sentence Transformers* | arcanumverse.org

- Built and operated a full-stack CTF platform serving 120+ users, sustaining 90+ concurrent connections and processing 12,000+ submissions with sub-second response times during a campus-wide event.
- Owned 3 of 10 trials end-to-end and built a real-time semantic scoring workflow using Sentence Transformers and cosine similarity for continuous answer-proximity feedback.

Canvas RAG System | *Python, ChromaDB, BM25, Sentence Transformers, RAGAS*

- Built a hybrid RAG pipeline for Canvas course content using ChromaDB dense retrieval, BM25 sparse search, query expansion, reciprocal-rank fusion, and CrossEncoder reranking.
- Achieved 0.95 faithfulness, 0.90 context recall, and 0.84 context precision in RAGAS evaluation of the hybrid retrieval and generation pipeline.